Practice Problems 2: Lectures 7–11

BUS 411 — Fixed Income Security Analysis and Valuation

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Instructions

These 20 numerical problems review Lectures 7–11. Four problems are provided for each lecture. Several are variants of Assignment 2 questions, while others extend worked examples from the lecture notes. Attempt each problem before opening its solution. Keep rates in decimal form during calculations and state monetary units.

1 Lecture 7: Interest-Rate Models

1.1 Problem 1: Historical Short-Rate Volatility

Weekly short-rate changes, in basis points, are

$$2,\ -2,\ 4,\ -1,\ 3,\ -3,\ 5,\ 0.$$

a. Compute the sample mean.

b. Compute the sample weekly standard deviation.

c. Annualize the standard deviation using $\sqrt{52}$.

d. Interpret the annualized result and identify one limitation.

Solution

There are eight observations. Their sum is 8 bps, so

$$\bar{x} = \frac{8}{8} = \boxed{1.00 \text{ bp per week}}.$$

The deviations from the mean are $1, -3, 3, -2, 2, -4, 4, -1$. Their squared deviations sum to

$$1 + 9 + 9 + 4 + 4 + 16 + 16 + 1 = 60.$$

Using the sample denominator $n - 1 = 7$,

$$s = \sqrt{\frac{60}{7}} = \boxed{2.9277 \text{ bps per week}}.$$

The square-root-of-time annualization is

$$s_{\text{annual}} = 2.9277\sqrt{52} = \boxed{21.11 \text{ bps per year}}.$$

This describes the annualized scale of short-rate changes if weekly changes are independent and have stable variance. Those assumptions may fail when volatility clusters or the monetary-policy regime changes.

1.2 Problem 2: Mean Reversion and a One-Standard-Deviation Shock

The short rate follows the Euler approximation

$$\Delta r = a(\theta - r)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon,$$

where $r = 5.00\%$, $\theta = 3.50\%$, $a = 0.80$, $\sigma = 1.20\%$ per year, $\Delta t = 0.25$, and $E[\varepsilon] = 0$.

a. Compute the expected quarterly rate change.

b. Compute the expected rate after one quarter.

c. Compute the standard deviation of the quarterly shock.

d. Give the rates corresponding to shocks of $\varepsilon = +1$ and -1.

Solution

The expected shock is zero. The mean-reversion component is

$$E[\Delta r] = 0.80(0.035 - 0.050)(0.25) = -0.0030 = -30 \text{ bps}.$$

Therefore,

$$E[r_{t+0.25}] = 0.0500 - 0.0030 = \boxed{4.70\%}.$$

The shock standard deviation is

$$\sigma\sqrt{\Delta t} = 0.012\sqrt{0.25} = 0.0060 = 60 \text{ bps}.$$

For $\varepsilon = +1$, the rate is $4.70\% + 0.60\% = \boxed{5.30\%}$; for $\varepsilon = -1$, it is $4.70\% - 0.60\% = \boxed{4.10\%}$. Mean reversion controls the center of the distribution, while σ controls its dispersion.

1.3 Problem 3: Nodes in a Lognormal Short-Rate Tree

A model specifies year-2 short rates as

$$r_{2,j} = a_2 e^{(2j-2)\sigma}, \qquad j = 0, 1, 2,$$

with $a_2 = 4.00\%$ and $\sigma = 0.15$.

a. Compute all three year-2 short rates.

b. Compute the one-year discount factor $1/(1 + r)$ at each node.

c. Which parameter sets the middle level of the rates?

d. If σ rises while a_2 is unchanged, what happens numerically to the distance between the low and high nodes?

Solution

For $j = 0, 1, 2$, the exponents are $-0.30, 0, +0.30$. Thus,

$$r_{2,0} = 0.04e^{-0.30} = \boxed{2.9633\%},$$
$$r_{2,1} = 0.04 = \boxed{4.0000\%},$$
$$r_{2,2} = 0.04e^{0.30} = \boxed{5.3994\%}.$$

The associated one-year discount factors are approximately

$$\boxed{0.9712, \quad 0.9615, \quad 0.9488}.$$

The level parameter a_2 fixes the middle node. The current low-to-high distance is $5.3994\% - 2.9633\% = 2.4361\%$. A larger σ lowers the low node and raises the high node, widening this distance; this is how volatility controls dispersion.

1.4 Problem 4: A One-Period Bond-Option Tree

One year from today, a zero-coupon bond will be worth either 94 in the up-rate state or 98 in the down-rate state. A European call expiring then has strike 96. The risk-neutral probability of the up-rate state is 0.55, and today's one-year short rate is 4%.

a. Compute the call payoff in each state.

b. Compute the call's time-0 value.

c. Compute the hedge ratio $(C_d - C_u)/(B_d - B_u)$.

d. Explain why the call pays more in the down-rate state.

Solution

The payoffs are

$$C_u = \max(94 - 96, 0) = 0, \qquad C_d = \max(98 - 96, 0) = 2.$$

Risk-neutral valuation gives

$$C_0 = \frac{0.55(0) + 0.45(2)}{1.04} = \boxed{0.8654}.$$

The option changes by 2 when the bond changes by 4, so

$$h = \frac{2 - 0}{98 - 94} = \boxed{0.50}.$$

When rates fall, discount factors rise and the zero-coupon bond price is higher. The call is therefore in the money in the down-rate state.

2 Lecture 8: Bonds with Embedded Options and OAS

2.1 Problem 5: Estimating a Static Spread

A corporate bond trades at 101.00. Discounting its promised cash flows at Treasury zero rates plus a trial spread gives:

Spread	Model value
40 bps	101.40
60 bps	100.72

a. Explain why the value falls as the spread rises.

b. Estimate the static spread by linear interpolation.

c. How far is the estimate from each trial spread?

d. State what risk the static spread does not model for a callable bond.

Solution

A larger spread raises every discount rate, reducing present value. Since 101.00 lies between the two trial values, the spread lies between 40 and 60 bps:

$$s \approx 40 + \frac{101.40 - 101.00}{101.40 - 100.72}(60 - 40) = \boxed{51.76 \text{ bps}}.$$

It is 11.76 bps above 40 and 8.24 bps below 60. Interpolation is an approximation because price is convex rather than perfectly linear in spread. A static spread also ignores the path-dependent exercise of an embedded call; OAS uses a rate lattice to model that behavior.

2.2 Problem 6: Pricing a Callable Bond

A two-year annual-pay bond has face value 100 and coupon 6. The time-0 short rate is 5%. At year 1, rates are either 4% or 8%, each with risk-neutral probability 0.5. The bond may be called at 101 at year 1, after the continuation value is calculated. Coupons are added after discounting the next payment.

a. Compute the year-2 payment.

b. Compute the year-1 option-free hold value at each node.

c. Apply the call rule and compute the time-0 callable value.

d. Compute the option-free value and the embedded call value.

Solution

The maturity payment is $100 + 6 = 106$. The year-1 hold values are

$$V_{1,L} = \frac{106}{1.04} + 6 = 107.9231,$$

$$V_{1,H} = \frac{106}{1.08} + 6 = 104.1481.$$

Both exceed the call price, so the issuer calls in both states and the ex-coupon node value is capped at 101. The callable value is

$$V_0^{\text{call}} = \frac{0.5(101) + 0.5(101)}{1.05} + 6 = \boxed{102.1905}.$$

Without the call,

$$V_0^{\text{straight}} = \frac{0.5(107.9231) + 0.5(104.1481)}{1.05} + 6 = \boxed{106.9863}.$$

Thus, the issuer's embedded call is worth

$$106.9863 - 102.1905 = \boxed{4.7958}$$

to the issuer and reduces the value received by the investor by the same amount.

2.3 Problem 7: Pricing a Putable Bond

A two-year annual-pay bond has face value 100 and coupon 3. The time-0 short rate is 5%. At year 1, rates are either 3% or 10%, each with probability 0.5. The investor may put the bond at 100 at year 1.

a. Compute the year-2 payment.

b. Compute the year-1 hold values.

c. Apply the put rule.

d. Compute the time-0 putable and option-free values and the embedded put value.

Solution

The maturity payment is $100 + 3 = 103$. The hold values are

$$V_{1,L} = \frac{103}{1.03} + 3 = 103.0000,$$

$$V_{1,H} = \frac{103}{1.10} + 3 = 96.6364.$$

The investor keeps the bond in the low-rate state and exercises the put in the high-rate state. Hence the year-1 values are 103 and 100. The time-0 putable value is

$$V_0^{\text{put}} = \frac{0.5(103) + 0.5(100)}{1.05} + 3 = \boxed{99.6667}.$$

The option-free value is

$$V_0^{\text{straight}} = \frac{0.5(103) + 0.5(96.6364)}{1.05} + 3 = \boxed{98.0649}.$$

The embedded investor put is worth

$$99.6667 - 98.0649 = \boxed{1.6017}.$$

2.4 Problem 8: OAS, Effective Duration, and Convexity

A callable bond has market price $P_0 = 100.00$. Trial OAS prices are 100.40 at 45 bps and 99.95 at 60 bps. After solving for OAS, the model gives $P_- = 102.60$ when the curve falls 50 bps and $P_+ = 97.70$ when it rises 50 bps.

a. Estimate the OAS.

b. Compute effective duration.

c. Compute effective convexity.

d. Interpret the two measures.

Solution

Interpolating to the market price,

$$\text{OAS} \approx 45 + \frac{100.40 - 100.00}{100.40 - 99.95}(60 - 45) = \boxed{58.33 \text{ bps}}.$$

With $\Delta y = 0.005$,

$$D_{\text{eff}} = \frac{P_- - P_+}{2P_0\Delta y} = \frac{102.60 - 97.70}{2(100)(0.005)} = \boxed{4.90}.$$

$$C_{\text{eff}} = \frac{P_- + P_+ - 2P_0}{P_0(\Delta y)^2} = \frac{102.60 + 97.70 - 200}{100(0.005)^2} = \boxed{120.00}.$$

Effective duration measures the local price slope after allowing the call decision to change. Effective convexity measures curvature under the same full repricing; both are therefore preferable to cash-flow duration for an option-embedded bond.

3 Lecture 9: Interest-Rate Futures

3.1 Problem 9: Treasury Futures Quotation and Margin Cash Flows

A trader buys one Treasury futures contract at 101-12. Subsequent settlements are 101-20, 101-08, and 101-24. Quotes are in 32nds, and one point is worth $\$1,000$.

a. Convert all four prices to decimals.

b. Compute each daily cash flow to the long.

c. Compute cumulative profit or loss.

d. Explain the sign of the cash flows.

Solution

The decimal prices are

$$101.375, \quad 101.625, \quad 101.250, \quad 101.750.$$

Daily marking to market gives

Day	Price change	Long cash flow
1	$+0.250$	$+\$250$
2	-0.375	$-\$375$
3	$+0.500$	$+\$500$

The cumulative result is $250 - 375 + 500 = \boxed{\$375}$, equal to $(101.750 - 101.375)(\$1{,}000)$. A long receives cash when the settlement price rises and pays cash when it falls.

3.2 Problem 10: Three-Month SOFR Futures

A long Three-Month SOFR futures position moves from 95.18 to 95.26. Each one-basis- point price move is worth $25 per contract.

a. Compute both implied SOFR rates.

b. Compute the price move in basis points.

c. Compute the long's dollar gain or loss.

d. Explain the inverse price-rate relation.

Solution

The quoted rate is 100− futures price. Initially it is $100 - 95.18 = \boxed{4.82\%}$ and afterward it is $100 - 95.26 = \boxed{4.74\%}$. The price rose by 0.08, or 8 basis points, so the long earns

$$8(\$25) = \boxed{\$200}.$$

The contract is quoted as 100 minus the implied rate; therefore, an increase in price mechanically corresponds to a decrease in the implied rate.

3.3 Problem 11: Cheapest-to-Deliver Bond

A Treasury futures contract settles at 112.50. Ignore accrued interest and financing.

Bond	Cash price	Conversion factor
A	119.10	1.0550
B	111.25	0.9850
C	106.40	0.9450

a. Compute each invoice price.

b. Compute each net delivery cost.

c. Identify the cheapest-to-deliver bond.

d. Explain why cash price alone is insufficient.

Solution

Invoice price equals futures price times conversion factor:

Bond	Invoice price	Net delivery cost: cash minus invoice
A	$112.50(1.055) = 118.6875$	0.4125
B	$112.50(0.985) = 110.8125$	0.4375
C	$112.50(0.945) = 106.3125$	0.0875

Bond C is the CTD because it has the smallest net delivery cost. Although C also has the lowest cash price here, the decision is based on cash price relative to the conversion-factor-adjusted invoice price, not cash price by itself.

3.4 Problem 12: Futures DV01 and a Portfolio Hedge

Using Bond C from Problem 11, assume modified duration 7.10, clean price 106.40, deliverable face value $\$100,000$, and conversion factor 0.9450. A long bond portfolio has DV01 of $\$30,500$.

a. Compute the CTD market value.

b. Compute CTD DV01.

c. Compute approximate futures DV01.

d. Compute the contract hedge and state its direction.

Solution

The deliverable market value is

$$MV_{CTD} = 1.0640(\$100{,}000) = \$106{,}400.$$

Therefore,

$$DV01_{CTD} = 7.10(\$106{,}400)(0.0001) = \boxed{\$75.5440}.$$

Adjusting for the conversion factor,

$$DV01_F \approx \frac{75.5440}{0.9450} = \boxed{\$79.9407}.$$

The hedge requires

$$N = \frac{30{,}500}{79.9407} = 381.53.$$

The manager should sell approximately $\boxed{382}$ futures contracts because the long bond portfolio loses when rates rise, while short futures gain. Rounding to a whole contract leaves a small residual DV01.

4 Lecture 10: Credit Exposure and Credit Risk

4.1 Problem 13: Spread Duration and DTS

A corporate bond position has market value $\$5,000,000$, spread duration 4.20, and spread 220 bps. Its spread widens by 30 bps.

a. Estimate the percentage price change.

b. Estimate the dollar change.

c. Compute DTS using spread in percentage points.

d. Interpret DTS.

Solution

Using $\Delta P/P \approx -SD\,\Delta s$ and $\Delta s = 0.0030$,

$$\frac{\Delta P}{P} \approx -4.20(0.0030) = \boxed{-1.26\%}.$$

The dollar change is

$$-0.0126(\$5{,}000{,}000) = \boxed{-\$63{,}000}.$$

With spread expressed as 2.20 percentage points,

$$DTS = 4.20(2.20) = \boxed{9.24}.$$

Equivalently, using decimal spread gives 0.0924. DTS combines sensitivity and the current spread level, providing a rough relative measure of spread risk across bonds with different credit quality.

4.2 Problem 14: Empirical Duration Multiplier

A \$20 million corporate bond position has analytical duration 6.50 and spread $s = 2.50\% = 0.025$. The correlation between Treasury-yield changes and relative spread changes is -0.60. Their volatilities are $\sigma_y = 0.25\%$ and $\sigma_s = 2.00\%$.

a. Compute $\beta = \rho\sigma_s/\sigma_y$.

b. Compute $M = 1 + s\beta$ and empirical duration.

c. Estimate the loss for a 50 bp Treasury-yield increase.

d. Compare it with the analytical-duration loss.

Solution

$$\beta = -0.60\frac{0.0200}{0.0025} = \boxed{-4.80}.$$

The duration multiplier and empirical duration are

$$M = 1 + 0.025(-4.80) = \boxed{0.88},$$

$$D_{emp} = 6.50(0.88) = \boxed{5.72}.$$

For $\Delta y = 0.0050$, the empirical estimate is

$$\frac{\Delta P}{P} \approx -5.72(0.005) = -2.86\%,$$

$$\Delta P = -0.0286(\$20{,}000{,}000) = \boxed{-\$572{,}000}.$$

Analytical duration gives $-6.50(0.005) = -3.25\%$, or $-\$650,000$. It overstates this empirically adjusted loss by $\boxed{\$78,000}$ because spreads are assumed to tighten when Treasury yields rise.

4.3 Problem 15: A One-Period Structural Credit Model

A firm owes \$100 million in one year. Its asset value will be \$140 million with probability 0.65 or \$70 million with probability 0.35. Ignore discounting.

a. Identify the default state.

b. Compute debt and equity payoffs in each state.

c. Compute expected debt and equity payoffs.

d. Compute expected debt loss and recovery in the default state.

Solution

Default occurs when assets are below promised debt, so it occurs in the \$70 million state.

Asset value	Debt payoff $\min(V_A, 100)$	Equity payoff $\max(V_A - 100, 0)$
$\$140m$	$\$100m$	$\$40m$
$\$70m$	$\$70m$	0

Expected payoffs are

$$E[D] = 0.65(100) + 0.35(70) = \boxed{\$89.5\text{m}},$$

$$E[E] = 0.65(40) + 0.35(0) = \boxed{\$26.0\text{m}}.$$

Expected debt loss relative to the promised \$100 million is \$10.5 million. Recovery conditional on default is $70/100 = \boxed{70\%}$, so loss given default is 30%. Equity is a call option on firm assets with strike equal to debt face value.

4.4 Problem 16: Default Intensity and Credit Spread

A reduced-form model has constant annual intensity $\lambda = 2.5\%$ and loss given default of 60%.

a. Compute one-year survival and default probabilities.

b. Compute five-year survival and cumulative default probabilities.

c. For 100 surviving issuers, compute expected defaults during the next year.

d. Estimate the credit spread using $s \approx \lambda \times LGD$.

Solution

For constant intensity, $S(t) = e^{-\lambda t}$. At one year,

$$S(1) = e^{-0.025} = 0.97531,$$

$$PD(1) = 1 - S(1) = \boxed{2.4690\%}.$$

At five years,

$$S(5) = e^{-0.125} = \boxed{88.2497\%},$$

$$PD(5) = 1 - S(5) = \boxed{11.7503\%}.$$

Among 100 initially surviving issuers, expected one-year defaults are $100(0.024690) = \boxed{2.469}$. The spread approximation is

$$s \approx 0.025(0.60) = 0.0150 = \boxed{150 \text{ bps}}.$$

An observed spread may differ because it also reflects liquidity, risk premia, taxes, and uncertainty about recovery and intensity.

5 Lecture 11: Interest-Rate Options

5.1 Problem 17: Treasury Futures Put Quotation

A trader buys one 110 Treasury futures put for 0-48. Option premiums are quoted in 64ths, and one point is worth $\$1,000$.

a. Convert the premium to points and dollars.

b. Compute the expiration break-even price in decimal and fractional notation.

c. Compute payoff and net profit if futures finish at 107-32.

d. Compute payoff and net profit if futures finish at 112-00.

Solution

The quote means 48/64 of one point:

$$0\text{-}48 = \frac{48}{64} = 0.75 \text{ point},$$

so the premium is $0.75(\$1,000) = \boxed{\$750}$. Break-even for a long put is

$$K - p = 110 - 0.75 = 109.25 = \boxed{109\text{-}16},$$

because $0.25 = 16/64$. At 107-32, $F_T = 107.50$, so

$$\text{payoff} = 110 - 107.50 = 2.50 \text{ points} = \$2,500,$$

$$\text{net profit} = 2,500 - 750 = \boxed{\$1,750}.$$

At 112-00 the put is out of the money, payoff is zero, and net profit is $\boxed{-\$750}$. It should not be exercised.

5.2 Problem 18: Put-Call Parity for Futures Options

European futures options have $F_0 = 112$, $K = 110$, $r = 4\%$, $T = 0.5$, and call premium $c = 3.10$ points.

a. Compute e^{-rT}.

b. Use $c - p = e^{-rT}(F_0 - K)$ to compute the fair put.

c. If the market put is 1.40, compute the mispricing in points and dollars.

d. Describe an arbitrage that locks in the difference.

Solution

$$e^{-rT} = e^{-0.04(0.5)} = \boxed{0.980199}.$$

Therefore,

$$p = c - e^{-rT}(F_0 - K) = 3.10 - 0.980199(2) = \boxed{1.1396 \text{ points}}.$$

The market put at 1.40 is overpriced by

$$1.40 - 1.1396 = 0.2604 \text{ point} = \boxed{\$260.40}.$$

At market prices, a long call and short put cost $3.10 - 1.40 = 1.70$ points and have expiration payoff $F_T - K$. Add a short futures contract, whose payoff is $F_0 - F_T$, and the combined expiration payoff is the certain amount $F_0 - K = 2$. Borrow its present value, 1.9604, to finance the 1.70 option cost. The time-0 surplus is $1.9604 - 1.70 = 0.2604$ point, while the expiration cash flows repay the borrowing.

5.3 Problem 19: Black-Scholes on a Zero-Coupon Bond

A European call and put are written on a zero-coupon bond with $S_0 = 95$, $K = 94$, $T = 0.75$, $r = 3\%$, and bond-price volatility $\sigma = 12\%$.

a. Compute $\sigma\sqrt{T}$, d_1, and d_2.

b. Use $N(d_1) = 0.6444$ and $N(d_2) = 0.6050$ to compute the call.

c. Compute the put using complementary normal probabilities.

d. Verify put-call parity.

Solution

$$\sigma\sqrt{T} = 0.12\sqrt{0.75} = \boxed{0.103923}.$$

$$d_1 = \frac{\ln(95/94) + (0.03 + 0.5(0.12)^2)(0.75)}{0.103923} = \boxed{0.3703},$$

$$d_2 = d_1 - 0.103923 = \boxed{0.2664}.$$

The call value is

$$c = 95(0.6444) - 94e^{-0.03(0.75)}(0.6050) = \boxed{5.61}.$$

Using $N(-d_1) = 0.3556$ and $N(-d_2) = 0.3950$,

$$p = 94e^{-0.0225}(0.3950) - 95(0.3556) = \boxed{2.52}.$$

The values satisfy parity up to rounding:

$$c - p \approx 5.61 - 2.52 = 3.09,$$

$$S_0 - Ke^{-rT} = 95 - 94e^{-0.0225} = 3.09.$$

5.4 Problem 20: Delta-DV01 Protective Put Hedge

A \$12 million bond portfolio has modified duration 7.50. A Treasury futures contract has DV01 \$75, current price 112, and point value \$1,000. A put has strike 111, premium 1.10 points, and delta -0.45.

a. Compute portfolio DV01 and futures-equivalent contracts.

b. Compute the number of puts for an initial delta-DV01 hedge and premium cost.

c. If rates rise 50 bps and futures fall 3.75 points at expiration, compute the portfolio loss, put payoff, and net hedged P&L using a whole-number hedge.

d. If rates instead fall 50 bps and the put expires worthless, compute net P&L.

Solution

Portfolio DV01 is

$$DV01_P = \$12{,}000{,}000(7.50)(0.0001) = \boxed{\$9,000}.$$

The portfolio is equivalent to

$$N_{fut-eq} = \frac{9,000}{75} = \boxed{120 \text{ futures contracts}}.$$

Because each put supplies only 0.45 short futures equivalent initially,

$$N_p = \frac{9,000}{0.45(75)} = 266.67,$$

so use 267 puts. Premium cost is

$$267(1.10)(\$1,000) = \boxed{\$293,700}.$$

For a 50 bp rate rise, the duration loss is

$$-\$9,000(50) = -\$450,000.$$

The expiration futures price is $112 - 3.75 = 108.25$. Each put pays $111 - 108.25 = 2.75$ points, so total payoff is

$$267(2.75)(\$1,000) = \$734,250.$$

Net hedged P&L is

$$-450,000 + 734,250 - 293,700 = \boxed{-\$9,450}.$$

If rates fall 50 bps, the mapped portfolio gain is $\$450,000$. With no put payoff,

$$\text{net P\&L} = 450,000 - 293,700 = \boxed{\$156,300}.$$

The hedge is delta-neutral only for a small move when initiated. By expiration the put's nonlinear payoff can create gains or losses different from a futures hedge, and the premium is paid in either scenario.